

Installation and Commissioning Quality Control Record

Welding visual inspection

Q-0400-011

ITP step #: 6300-3.3

Revision # 00

Project: GMS

Unit : 4

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Visual rotor rib after welding

Applicable standard

- ☒ CSA W59 Article 5.9
☐ ANSI/AWS D1.1 Article 3.6
☐ ASME B31.1 article 136.4
 (piping)

Welding joint inspected	ITP step #	Any superficial cracks	Any lack of fusion between welding and the base metal	Any crater	Welding profiles acceptable according to standard	Dimension in conformity with drawing	Date (dd-mmm-yyyy)	Initials
Rotor Rib #1	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #2	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #3	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #4	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #5	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #6	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #7	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #8	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #9	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #10	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #11	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #12	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #13	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #14	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #15	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM

Comments : _____

NCR #: _____

Measurement Instruments	T°	Measured by: _____	dd-mmm-yyyy
Visual & fillet gauge	Before: _____	_____	_____
_____	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>28-May-2012</u>
_____	_____	Cust. Repres. _____	_____

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Applicable standard

- ☒ CSA W59 Article 5.9
☐ ANSI/AWS D1.1 Article 3.6
☐ ASME B31.1 article 136.4
 (piping)

Welding joint inspected	ITP step #	Any superficial cracks	Any lack of fusion between welding and the base metal	Any crater	Welding profiles acceptable according to standard	Dimension in conformity with drawing	Date (dd-mmm-yyyy)	Initials
Rotor Rib #16	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #17	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #18	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #19	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM
Rotor Rib #20	6300-3.3	No	No	No	Yes	Yes	28-May-2012	LM

Comments : _____

NCR #: _____

Measurement Instruments	T°	Measured by: _____	dd-mmm-yyyy
Visual & fillet gauge	Before: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>28-May-2012</u>
	After: _____	Cust. Repres. _____	_____

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Visual rotor gussets after welding

Applicable standard

- ☒ CSA W59 Article 5.9
☐ ANSI/AWS D1.1 Article 3.6
☐ ASME B31.1 article 136.4
 (piping)

Welding joint inspected	ITP step #	Any superficial cracks	Any lack of fusion between welding and the base metal	Any crater	Welding profiles acceptable according to standard	Dimension in conformity with drawing	Date (dd-mmm-yyyy)	Initials
Rotor Gusset #1	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #2	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #3	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #4	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #5	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #6	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #7	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #8	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #9	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #10	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #11	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #12	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #13	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #14	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #15	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM

Comments : _____

NCR #: _____

Measurement Instruments	T°	Measured by: _____	dd-mmm-yyyy
Visual & fillet gauge	Before: _____	_____	_____
_____	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>4-Jun-2012</u>
_____	_____	Cust. Repres. _____	_____

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Visual rotor gusset after welding

Applicable standard

- ☒ CSA W59 Article 5.9
- ☐ ANSI/AWS D1.1 Article 3.6
- ☐ ASME B31.1 article 136.4
(piping)

Welding joint inspected	ITP step #	Any superficial cracks	Any lack of fusion between welding and the base metal	Any crater	Welding profiles acceptable according to standard	Dimension in conformity with drawing	Date (dd-mmm-yyyy)	Initials
Rotor Gusset #16	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #17	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #18	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #19	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM
Rotor Gusset #20	6300-3.4	No	No	No	Yes	Yes	4-Jun-2012	LM

Comments : _____

NCR #: _____

Measurement Instruments	T°	Measured by: _____	dd-mmm-yyyy
Visual & fillet gauge	Before: _____	_____	_____
_____	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>4-Jun-2012</u>
_____	_____	Cust. Repres. _____	_____

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after welding

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
Gusset #1 Joint I1	6300-3.4	X					03-06-12	LM
Gusset #1 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #1 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #1 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #1 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #1 Joint O3	6000-3.4	x					03-06-12	LM
Gusset #2 Joint I1	6000-3.4	x					03-06-12	LM
Gusset #2 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #2 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #2 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #2 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #2 Joint O3	6000-3.4	x					03-06-12	LM
CONTINUED NEXT PAGE								

Comments : No indications were observed.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>4-Jun-2012</u>
		Cust. Repres. _____	

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after welding

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
Gusset #3 Joint I1	6300-3.4	X					03-06-12	LM
Gusset #3 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #3 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #3 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #3 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #3 Joint O3	6000-3.4	x					03-06-12	LM
Gusset #4 Joint I1	6000-3.4	x					03-06-12	LM
Gusset #4 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #4 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #4 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #4 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #4 Joint O3	6000-3.4	x					03-06-12	LM
CONTINUED NEXT PAGE								

Comments : No indications were observed.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>4-Jun-2012</u>
		Cust. Repres. _____	

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after welding

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
Gusset #5 Joint I1	6300-3.4	X					03-06-12	LM
Gusset #5 Joint I2	6000-3.4	X					03-06-12	LM
Gusset #5 Joint I3	6000-3.4	X					03-06-12	LM
Gusset #5 Joint O1	6000-3.4	X					03-06-12	LM
Gusset #5 Joint O2	6000-3.4	X					03-06-12	LM
Gusset #5 Joint O3	6000-3.4	X					03-06-12	LM
Gusset #6 Joint I1	6000-3.4	X					03-06-12	LM
Gusset #6 Joint I2	6000-3.4	X					03-06-12	LM
Gusset #6 Joint I3	6000-3.4	X					03-06-12	LM
Gusset #6 Joint O1	6000-3.4	X					03-06-12	LM
Gusset #6 Joint O2	6000-3.4	X					03-06-12	LM
Gusset #6 Joint O3	6000-3.4	X					03-06-12	LM
CONTINUED NEXT PAGE								

Comments : No indications were observed.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>4-Jun-2012</u>
		Cust. Repres. _____	

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after welding

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
Gusset #7 Joint I1	6300-3.4	X					03-06-12	LM
Gusset #7 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #7 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #7 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #7 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #7 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #8 Joint I1	6000-3.4	x					03-06-12	LM
Gusset #8 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #8 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #8 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #8 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #8 Joint O3	6000-3.4	x					03-06-12	LM
CONTINUED NEXT PAGE								

Comments : No indications were observed.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>4-Jun-2012</u>
		Cust. Repres. _____	

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after welding

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
Gusset #9 Joint I1	6300-3.4	X					03-06-12	LM
Gusset #9 Joint I2	6000-3.4	X					03-06-12	LM
Gusset #9 Joint I3	6000-3.4	X					03-06-12	LM
Gusset #9 Joint O1	6000-3.4	X					03-06-12	LM
Gusset #9 Joint O2	6000-3.4	X					03-06-12	LM
Gusset #9 Joint O3	6000-3.4	X					03-06-12	LM
Gusset #10 Joint I1	6000-3.4	X					03-06-12	LM
Gusset #10 Joint I2	6000-3.4	X					03-06-12	LM
Gusset #10 Joint I3	6000-3.4	X					03-06-12	LM
Gusset #10 Joint O1	6000-3.4	X					03-06-12	LM
Gusset #10 Joint O2	6000-3.4	X					03-06-12	LM
Gusset #10 Joint O3	6000-3.4	X					03-06-12	LM
CONTINUED NEXT PAGE								

Comments : No indications were observed.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>4-Jun-2012</u>
		Cust. Repres. _____	

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after welding

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
Gusset #11 Joint I1	6300-3.4	X					03-06-12	LM
Gusset #11 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #11 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #11 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #11 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #11 Joint O3	6000-3.4	x					03-06-12	LM
Gusset #12 Joint I1	6000-3.4	x					03-06-12	LM
Gusset #12 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #12 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #12 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #12 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #12 Joint O3	6000-3.4	x					03-06-12	LM
CONTINUED NEXT PAGE								

Comments : No indications were observed.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>4-Jun-2012</u>
		Cust. Repres. _____	

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after welding

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
Gusset #13 Joint I1	6300-3.4	X					03-06-12	LM
Gusset #13 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #13 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #13 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #13 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #13 Joint O3	6000-3.4	x					03-06-12	LM
Gusset #14 Joint I1	6000-3.4	x					03-06-12	LM
Gusset #14 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #14 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #14 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #14 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #14 Joint O3	6000-3.4	x					03-06-12	LM
CONTINUED NEXT PAGE								

Comments : No indications were observed.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>4-Jun-2012</u>
		Cust. Repres. _____	

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after welding

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
Gusset #15 Joint I1	6300-3.4	X					03-06-12	LM
Gusset #15 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #15 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #15 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #15 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #15 Joint O3	6000-3.4	x					03-06-12	LM
Gusset #16 Joint I1	6000-3.4	x					03-06-12	LM
Gusset #16 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #16 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #16 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #16 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #16 Joint O3	6000-3.4	x					03-06-12	LM
CONTINUED NEXT PAGE								

Comments : No indications were observed.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>4-Jun-2012</u>
		Cust. Repres. _____	

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Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
Gusset #17 Joint I1	6300-3.4	X					03-06-12	LM
Gusset #17 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #17 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #17 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #17 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #17 Joint O3	6000-3.4	x					03-06-12	LM
Gusset #18 Joint I1	6000-3.4	x					03-06-12	LM
Gusset #18 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #18 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #18 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #18 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #18 Joint O3	6000-3.4	x					03-06-12	LM
CONTINUED NEXT PAGE								

Comments : No indications were observed.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
<u>Magnetic Particles</u>	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>4-Jun-2012</u>
		Cust. Repres. _____	

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Welding joint or part inspected	ITP step #	Test type					Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic	Default detected		
Gusset #19 Joint I1	6300-3.4	X					03-06-12	LM
Gusset #19 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #19 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #19 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #19 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #19 Joint O3	6000-3.4	x					03-06-12	LM
Gusset #20 Joint I1	6000-3.4	x					03-06-12	LM
Gusset #20 Joint I2	6000-3.4	x					03-06-12	LM
Gusset #20 Joint I3	6000-3.4	x					03-06-12	LM
Gusset #20 Joint O1	6000-3.4	x					03-06-12	LM
Gusset #20 Joint O2	6000-3.4	x					03-06-12	LM
Gusset #20 Joint O3	6000-3.4	x					03-06-12	LM

Comments : No indications were observed.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>4-Jun-2012</u>
		Cust. Repres. _____	

RADIUS

NDE SERVICES INC.

9004 - 107th Street
Fort St. John, BC V1J 5R5
(250) 787-0336 phone
(250) 787-0733 fax

Client:	Andritz Hydro Ltd	Date	12.06.04
Attention:	Jean-Luc Cote	Page	1 of 2
Charge Code:	Unit #4 Refurbishment	Ticket #:	14343
Location:	GM Shrum Generating Station		

MAGNETIC PARTICLE INSPECTION REPORT

TEST METHOD:	Colour Contrast / A/C Yoke
ACCEPTANCE STANDARD:	Client Spec / Crack Detection
OBJECT:	Pocket Gussets

TEST CONDITIONS

EQUIPMENT:	MANUFACTURER: Electro Spect				TYPE: ES-X	
MAGNETIC PARTICLES:	MANUFACTURER: Magnaflux				TYPE: 7HF	
	☐ DRY	☒ WET	☐ FLUORESCENT	☒ NON-FLUORESCENT COLOUR: Black		
MAGNETIC METHOD:	☐ PROD	☒ YOKE	☐ COIL	☒ CONTINUOUS	☐ RESIDUAL	
SURFACE CONDITION:	☒ AS GROUND	☐ MACHINED	☐ SHOT BLASTED	☐ BASE METAL	☐ PAINTED SURFACE	
					☐ AC ☐ DC ☐ Buffed	

EXAMINATION RESULTS:

Inspected six welds on each of 19 Pocket Gussets in the rotor for unit #4, one remaining Pocket Gusset wasn't installed at the time of inspection. The inspection consisted of wet, colour contrast, multi-directional A/C magnetic particle scans of the welded joints and approximately one inch of parent material on either side of the weld area in accordance with standard industry practice.

No relevant indications were noted at this time.

Please see the following photograph page.

Six indications on the Spider Arms identified in photographs # 1 thru 6 of the report dated May 09, 2012 were removed by grinding and repair welding operations with the exception of the light indications in photograph #2 which was removed by grinding only. All areas were then re-inspected to confirm the complete elimination of the indications, no relevant indications were noted at the time of inspection.

W. Fuhr

TECHNICIAN

SIGNATURE

II

12707

REGISTRATION #

The above examination results are the professional opinion of the Technician, but in no way guarantee the integrity of the product.

D. Beamish

TECHNICIAN

SIGNATURE

II

5380

REGISTRATION #

RADIUS

NDE SERVICES INC.

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PHOTOGRAPHS



Outer view of Pocket Gusset



Inner view of Pocket Gusset